

BSc (Hons) Computing - Coventry University
Higher National Diploma in Software Engineering - NIBM



Data Warehousing and Business Intelligence

Niranga Dharmaratna
niranga@vl.nibm.lk

Learning Outcomes



Course Aim/ Intended Learning Outcomes:

At the completion of this module student will be able to

1. Explain the concepts of Data warehousing and decision support systems.
2. Apply the de-normalized concept of data modeling in data warehousing.
3. Use of Oracle BI / MS Power BI tools for building Data Warehouses.
4. Explain the fundamentals of OLAP Cube creation.
5. Use data visualization for business intelligence.

Module Topics



- 6.1. Introduction to Data Warehousing, Evolution of Decision Support Systems
- 6.2. Data Modeling with Denormalized concepts
- 6.3. Data preprocessing, Data Cleansing, Data Integration, Data Transformation and Data Reduction - (ETL Process)
- 6.4. OLAP Cube creation with facts and dimensions
- 6.5. Introduction to Business Intelligence
- 6.6. Data Visualization

Learning Plan



Teaching	45 hours
Independent Learning	?

Prerequisites

For hands-on learning, practicals and coursework

- **Familiarity with Databases**
 - **Familiarity with SQL**
 - **Data Awareness - Rough idea about data and information**
- **Oracle Cloud Infrastructure (OCI) account**
 - **Oracle SQL Developer**
 - **Tableau Desktop - Tableau for Students**

Reference Material

Data Mining Concepts and Techniques
Jiawei Han, Micheline Kamber, Jian Pei
ISBN: 978-0123814791
Published by Morgan Kaufmann

The Data Warehouse ETL Toolkit
Ralph Kimball and Joe Caserta
ISBN: 978-0764567575
Published by Wiley & Sons, Incorporated

Module Evaluation



Module Evaluation Criteria:

- Coursework - 50%
- Exams (Closed book) - 50%

Further Points



Teaching:

- Sub-modules Order
- Importance of Lectures
- Notes / Terminology
- References

Coursework:

- Approaches and Evaluation
- Coursework Viva

Further Points

 ...Continued.

My Evaluation Policy:

- Credits for hard work. Hard means knowledge, innovation and skills.

My AI Generated Content Policy:

- SIMPLE. Should be able to defend your work.

Questions and Discussion